order and degree of differentail equations and linearity and non linearity of differential equations

Here is a clear and simple explanation of **order, degree, linearity, and non-linearity** of differential equations.

Differential Equations: Order, Degree, Linearity & Non-Linearity

1) Order of a Differential Equation

The **order** of a differential equation is the **highest order derivative** present in the equation.

Examples

1.

$$\frac{dy}{dx} + y = 0$$

Highest derivative: $\frac{dy}{dx}$

Order = 1

2.

$$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + y = 0$$

Highest derivative: $\frac{d^2y}{dx^2}$

Order = 2

3.

$$\frac{d^3y}{dx^3} + y = 5$$

Highest derivative: $\frac{d^3y}{dx^3}$

Order = 3

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2) Degree of a Differential Equation

The **degree** is the **power of the highest order derivative**, after the equation is made **free from radicals and fractions** involving derivatives.

Examples

1.

$$\left(\frac{dy}{dx}\right)^2 + y = 0$$

Highest order derivative: $\frac{dy}{dx}$

Power = 2

Degree = 2

2.

$$\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^3 = 0$$

Highest derivative: $\frac{d^2y}{dx^2}$

Power = 1

Degree = 1

3.

$$\sqrt{\frac{dy}{dx}} + y = 0$$

Remove radical:

$$\frac{dy}{dx} = y^2$$

Power of highest derivative = 1

Degree = 1

⚠️ If radicals or fractions of derivatives **cannot be removed**, degree is **not defined**.

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3) Linear Differential Equation

A differential equation is **linear** if:

- The dependent variable and its derivatives appear only in **first power**.
- No products between y and its derivatives.
- No functions like $y^2, \sin y, e^y$, etc.

General form (first order linear):

$$\frac{dy}{dx} + P(x)y = Q(x)$$

Examples

1.

$$\frac{dy}{dx} + y = 0$$

Linear.

2.

$$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + y = 5$$

Linear.

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4) Non-Linear Differential Equation

A differential equation is **non-linear** if:

- The dependent variable or derivatives appear in **powers other than 1**.
- There are **products** of y and its derivatives.
- Functions like $y^2, \sin y, e^y$ appear.

Examples

1.

$$\left(\frac{dy}{dx}\right)^2 + y = 0$$

Non-linear (derivative squared).

2.

$$\frac{dy}{dx} + y^2 = 0$$

Non-linear (y^2).

3.

$$y\frac{dy}{dx} + x = 0$$

Non-linear (product of y and derivative).

Quick Comparison Table

Feature	Linear Equation	Non-Linear Equation
Power of y	Only 1	More than 1 allowed
Product $y \cdot \frac{dy}{dx}$	Not allowed	Allowed
Functions of y	Not allowed	Allowed
Example	$dy/dx + y = 0$	$dy/dx + y^2 = 0$

Very Short Summary

- **Order** → Highest derivative.
- **Degree** → Power of highest derivative.
- **Linear** → No powers, no products, no functions of y.
- **Non-linear** → Any power, product, or function of y.

If you want, I can:

- Give **practice questions**.
- Provide **MCQs for exams**.
- Make a **one-page revision sheet** or **PDF**.